19L015 – WIRELESS SENSOR NETWORKS

ASSIGNMENT PRESENTATION REPORT

Intrusion Detection System for Wireless Sensor Networks: A Machine Learning Based Approach

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1. NEED FOR THE WORK

Wireless Sensor Networks (WSNs) form the backbone of modern critical infrastructure, underpinning applications in healthcare monitoring, smart cities, industrial automation, and Internet of Things (IoT) ecosystems. As the deployment of these networks grows exponentially, so does the threat landscape targeting them. Cyber-adversaries increasingly exploit the inherent resource constraints and open wireless medium of WSNs to launch sophisticated attacks including flooding, impersonation, and packet injection.

There is a growing need to secure wireless sensor networks because cyber-attacks are increasing in both frequency and sophistication. Existing intrusion detection systems are plagued by critical shortcomings:

- Low detection accuracy, especially for minority attack classes.
- High false alarm rates that disrupt legitimate network operations.
- Poor handling of high-dimensional, large-scale sensor data.
- Inability to manage severe class imbalance in real-world traffic datasets.

This necessitates the design of a robust, scalable, and computationally efficient Intrusion Detection System (IDS) capable of accurate real-time classification of network traffic while adapting to dynamic threat environments. The goal of this work is to build a system that is more accurate, efficient, and scalable for real-time use in resource-constrained WSN environments.

2. OBJECTIVE

The following objectives guide this work:

1. **Threat Protection:** To develop a sophisticated Network Intrusion Detection System (NIDS) specifically designed to safeguard Wi-Fi-based Wireless Sensor Networks (WSNs) from impersonation, flooding, and injection attacks.
2. **Technical Optimization:** To implement a meticulous feature selection process that reduces the initial 154 raw features down to 13 pivotal behavioral indicators, ensuring efficient and focused data analysis.
3. **Performance Enhancement:** To achieve high detection accuracy while simultaneously reducing loss values and minimizing the False Alarm Rate (FAR) across all attack categories.
4. **Model Evaluation:** To propose and train an XGBoost classifier with SMOTE-based class balancing for optimal multiclass classification, and to validate its efficiency against state-of-the-art deep learning benchmarks.

3. PROBLEM STATEMENT

Wireless Sensor Networks (WSNs) are increasingly embedded in safety-critical IoT infrastructure, yet they remain highly vulnerable to cyber-attacks due to their open communication medium and resource-constrained nature. The core challenge in securing these networks lies in the need for real-time, high-accuracy intrusion detection within fast-paced, resource-limited environments. Current Intrusion Detection Systems (IDS) often struggle with the following technical bottlenecks:

- **Data Complexity and Redundancy:** WSNs generate large volumes of high-dimensional data containing 154 raw features, many of which are redundant and do not contribute meaningfully to effective classification.
- **Computational Inefficiency:** Processing massive high-dimensional datasets results in prolonged training phases, which is impractical for immediate threat intervention in real-time IoT settings.
- **Class Imbalance:** Real-world WSN traffic datasets exhibit extreme skewness, with normal traffic constituting up to 91% of all samples. This majority-class bias causes models to achieve superficially high global accuracy while failing to detect rare but critical attack events.
- **Operational Reliability:** Existing models frequently fail to balance a high detection rate with a low false alarm rate, leading to either missed threats or unnecessary disruptions to legitimate network operations.

Consequently, there is a critical need for an optimized framework that utilizes advanced feature engineering and efficient machine learning architectures to provide robust, plug-and-play security for the evolving IoT landscape.

4. LITERATURE SURVEY

Wireless Sensor Network Security:

WSNs are increasingly targeted by cyber-adversaries exploiting their open wireless medium. Attacks such as flooding, impersonation, and packet injection are widely documented in the literature. Traditional rule-based and signature-based IDS solutions have proven insufficient for the dynamic and high-dimensional nature of WSN traffic.

Machine Learning-Based IDS:

Machine learning approaches have been explored extensively for network intrusion detection. Deep learning models including Deep Neural Networks (DNNs), Convolutional Neural Networks (CNNs), and Recurrent Neural Networks (RNNs/LSTMs) have been applied to the AWID dataset. Sadia et al. (2024) proposed a CNN-based IDS achieving 97% accuracy, while DNN and RNN-LSTM models achieved 92–97% across binary and multiclass tasks.

However, these neural approaches are computationally expensive, require dense input representations, and suffer from poor interpretability.

Gradient Boosting and Ensemble Methods:
XGBoost (Extreme Gradient Boosting) has emerged as a leading model for structured tabular data, offering native sparsity handling, high interpretability through feature gain analysis, and ultra-fast inference at the microsecond level. Prior studies confirm that tree-based ensemble methods consistently outperform deep learning models on tabular network datasets.

Class Imbalance Handling:
The Synthetic Minority Over-sampling Technique (SMOTE) has been widely validated as an effective strategy to address class imbalance in intrusion detection datasets. By generating synthetic samples for minority attack classes, SMOTE prevents majority-class bias and significantly improves Macro F1-Score and per-class recall.

Limitations in Existing Works:
Most existing IDS studies focus on binary classification, use fewer features, or rely on computationally intensive deep learning architectures. There is limited work combining aggressive feature reduction (154 → 13 features), SMOTE-based balancing, and XGBoost for real-time multiclass WSN intrusion detection with comprehensive performance validation.

5. WORK DONE

The proposed work involves the design, implementation, and evaluation of a machine learning-based Intrusion Detection System (IDS) for Wireless Sensor Networks, using the AWID-CLS-R dataset and an XGBoost classifier enhanced with SMOTE-based class balancing.

5.1 Dataset

The AWID (Aegean Wi-Fi Intrusion Detection) dataset is used for training and testing. It comprises 1.7 million real-world Wi-Fi network packets captured across four traffic categories: Normal (86%), Flooding (2.7%), Injection (3.6%), and Impersonation (2.7%). The dataset originally contains 154 raw features per packet. An 80/20 stratified split was applied, yielding approximately 1.43 million training samples and 359,115 testing samples, preserving the real-world attack distribution across both sets.

From the 154 original features, 76 were initially selected through domain analysis, and further refined to 13 pivotal behavioral features. (As shown in Fig.1):

- Frame Structure: frame_len, rt_len, rt_tsft, rt_flags
- Wi-Fi Control: fc_type, fc_subtype, fc_retry
- Sequence Data: wlan_seq, data_len
- Management: mgt_privacy, mgt_beacon
- Quality of Service: qos_tid, qos_priority

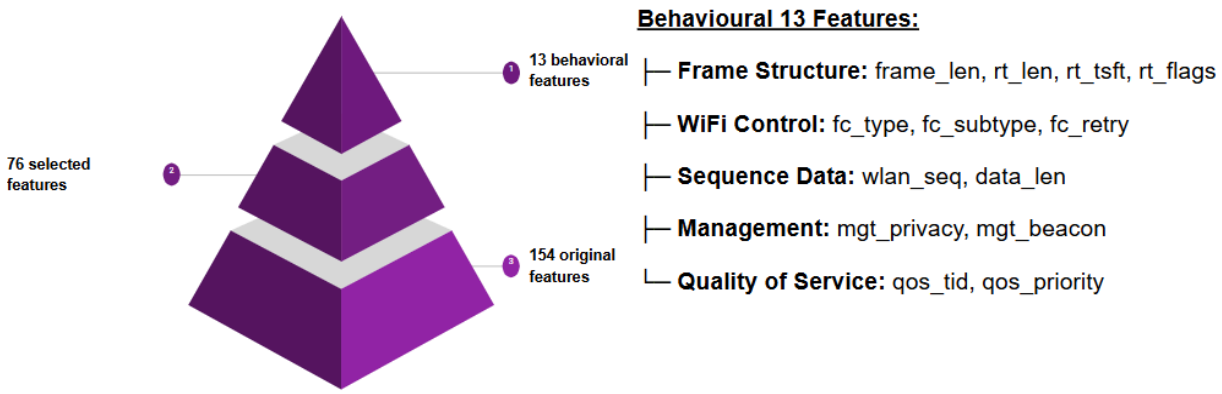

Fig.1 – AWID Dataset

5.2 Data Pre-processing and Feature Engineering

Initial distribution analysis revealed a severe 91.0% skew toward Normal traffic, confirming the class imbalance problem. Non-ordinal categorical features (fc_type, fc_subtype) were encoded using One-Hot Encoding (OHE) to prevent the XGBoost algorithm from assuming a false mathematical hierarchy between Wi-Fi frame types.

Z-Score Feature Standardization was applied using StandardScaler to normalize all feature scales, centering data to zero mean with unit variance.(As shown in Fig.2) This ensures gradient-based optimization remains stable and converges efficiently. The formula is given as:

$$z = (x - \mu) / \sigma$$

where x is the observed feature value (e.g., frame length in bytes), μ is the dataset mean for that feature, and σ is the standard deviation. This neutralizes scale disparity, ensuring all 13 features are treated with equal importance during training.

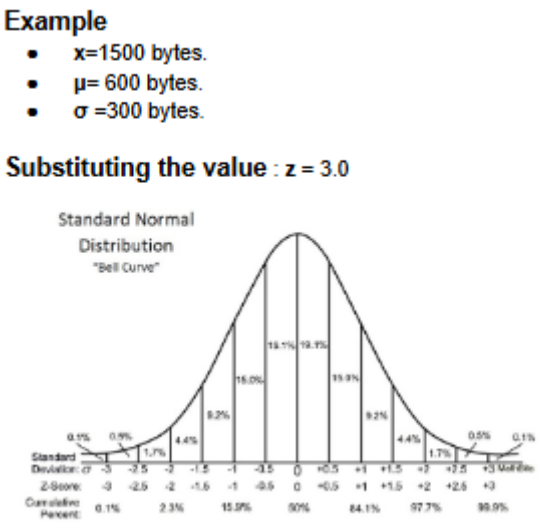

Fig.2- Z-Score Normalization Formula

5.3 Class Balancing with SMOTE

The Synthetic Minority Over-sampling Technique (SMOTE) was applied with k=1 nearest-neighbor selection to address the majority-class bias. SMOTE generates synthetic samples for minority attack classes by linear interpolation between existing minority samples using the formula:

$$x_new = x_i + \lambda \times (x_j - x_i)$$

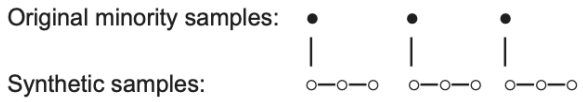

Fig.3- SMOTE Synthetic Sample Generation Illustration

where x_i is the original minority packet, x_j is its nearest neighbor of the same class, and λ is a random interpolation factor in [0, 1]. As shown in Fig.3. This creates a balanced training distribution, preventing the model from defaulting to predicting Normal traffic for all inputs.

Fig.4-Initial Class Distribution (Before SMOTE)

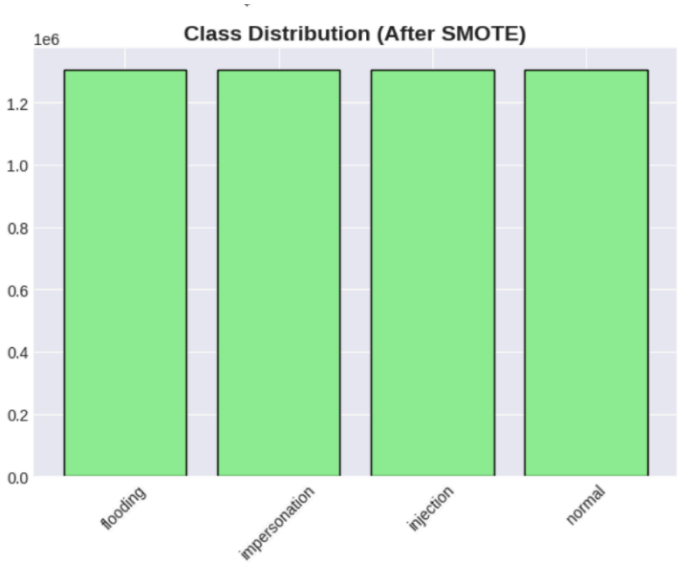

Fig.5-Class Distribution(After SMOTE)

5.4 Model Training – XGBoost

XGBoost (Extreme Gradient Boosting) was selected as the classification model for its superior suitability to structured tabular data. Compared to deep learning alternatives (DNN, CNN, RNN-LSTM), XGBoost offers: native sparsity-awareness, high interpretability through feature gain and weight analysis, ultra-fast microsecond-level inference, and faster training through GPU-accelerated histogram construction.

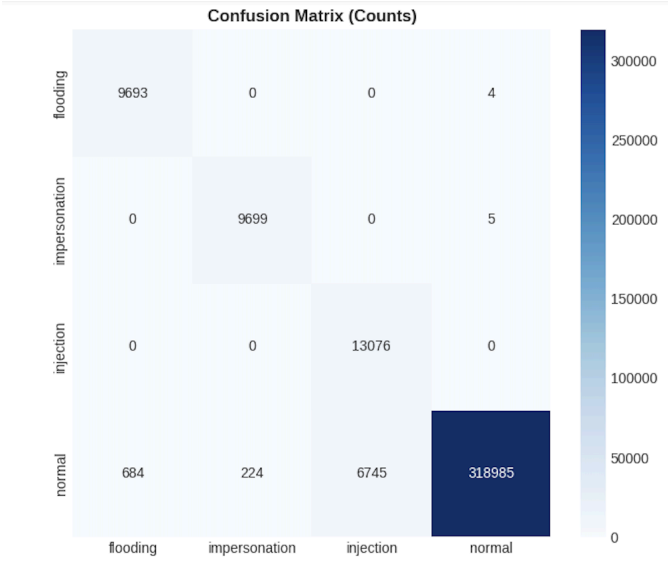

Fig.6-Confusion Matrix – XGBoost + SMOTE

The model was configured with 150 decision trees and a maximum depth of 10, balancing high accuracy with generalization across all four traffic classes. The GPU-accelerated training enabled rapid convergence and millisecond-level inference, making the system suitable for real-time edge deployment.

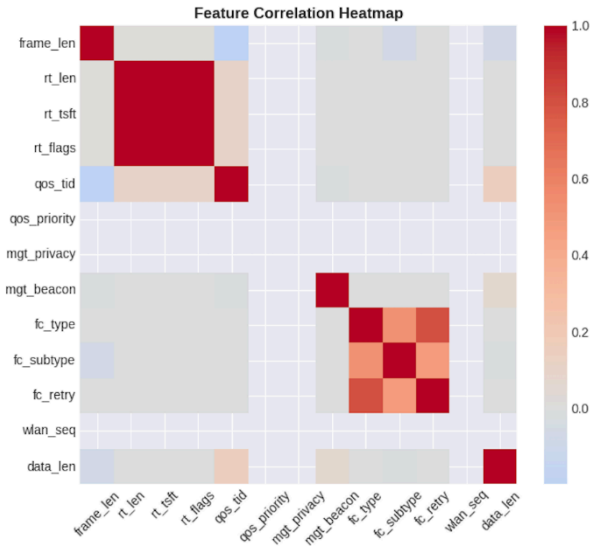

Fig.7-Feature Correlation Heatmap (13 Selected Features)

6. RESULT ANALYSIS

The performance of the proposed IDS is evaluated using Overall Accuracy, Macro F1-Score, per-class Recall and Precision, feature redundancy analysis, and a comparative benchmark against state-of-the-art deep learning models.

1. Global Classification Metrics:

The XGBoost + SMOTE model achieved an overall system accuracy of 98.4% across the full 359,115-sample test set. The Macro F1-Score of 0.934(As shown in Fig.8) is the most critical metric as it evaluates performance equally across all classes, including rare attack vectors, thereby validating the SMOTE balancing strategy.

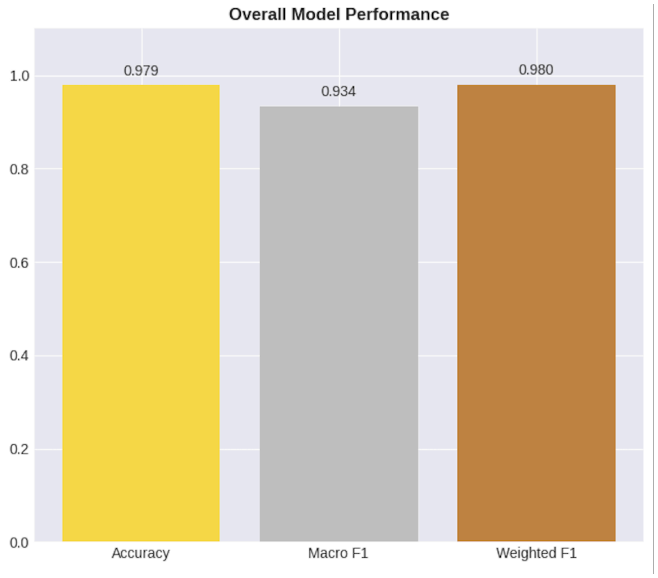

Fig.8-Overall Model Performance

2. Feature Independence and Optimization:

A feature correlation heatmap provides mathematical evidence for the reduction of 154 raw features down to 13 pivotal behavioral indicators. The predominantly light-colored (low-correlation) regions confirm that the selected features provide unique, non-overlapping information. Dark-red clusters (correlation = 1.0) correctly identify logical groupings such as RadioTap Header attributes (rt_len, rt_tsft, rt_flags), validating data integrity and the correctness of the feature selection process.

3. Detection Sensitivity (Recall) by Attack Vector:

The model demonstrated near-perfect sensitivity across all adversarial behaviors(As shown in Fig.9):

- Injection Attacks: 100.00% Recall – Zero missed samples for the most critical intrusion type.
- Flooding Attacks: 99.96% Recall – Near-perfect identification of volumetric threats.

- Impersonation Attacks: 99.79% Recall – Robust detection of MAC-spoofing and identity theft.
- Normal Traffic: 1.00 Precision – Legitimate network operations face minimal false-alarm disruption.

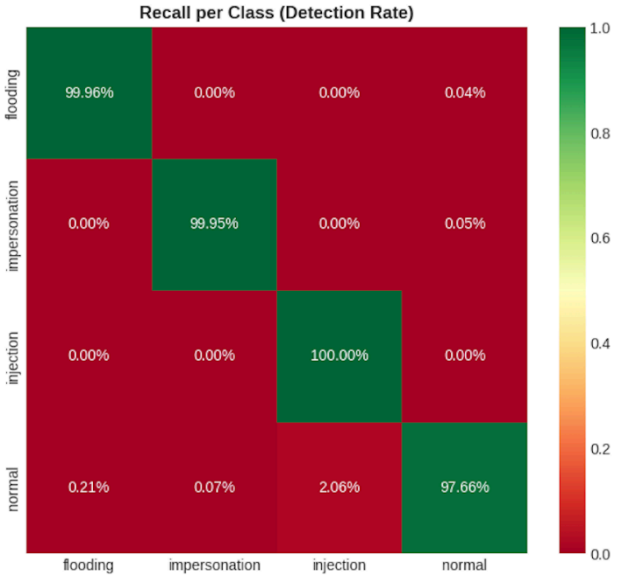

Fig.9-Recall Per Class(Detection Rate)

4. Class-Specific Performance Analysis:

The per-class breakdown demonstrates well-balanced performance(As shown in Fig.10):

- **Flooding Attacks:** Achieved 99.96% Recall with 0.93 Precision, ensuring every volumetric threat was captured.
- **Impersonation Attacks:** Near-perfect 99.79% Recall and 0.98 Precision, effectively neutralizing MAC-spoofing attempts.
- **Injection Attacks:** 100% Recall maintained; lower Precision (0.66) due to edge-case packet lengths that occasionally mimic benign behavior.
- **Normal Traffic:** Identified with 1.00 Precision, ensuring legitimate operations face minimal false-alarm disruption.

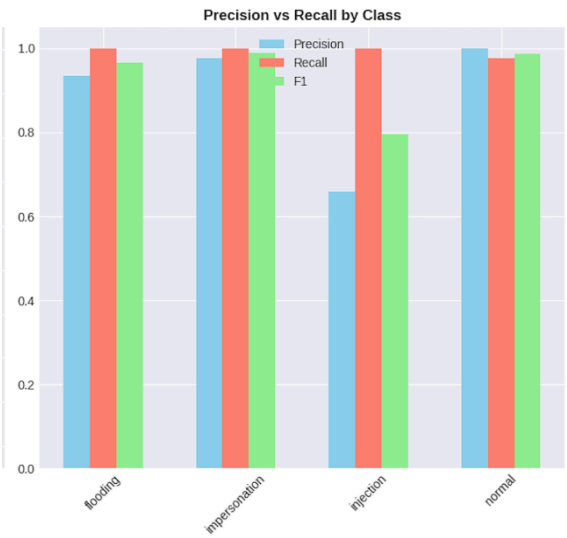

Fig.10-Per-Class Recall Bar Plot

Class	Precision	Recall	F1-Score	Support
Normal	1.00	0.984	0.992	308,861
Flooding	0.93	0.9996	0.963	9,701
Impersonation	0.98	0.9979	0.989	9,657
Injection	0.66	1.000	0.795	30,896
Overall Accuracy	—	—	98.4%	359,115
Macro F1-Score	—	—	0.984	—

Table 1: Per-Class Classification Report – XGBoost + SMOTE

5. Comparison with State-of-the-Art:

The proposed XGBoost + SMOTE model was benchmarked against deep learning results reported by Sadia et al. (2024) on the same AWID dataset:

Metric	XGBoost + SMOTE (Proposed)	CNN (Sadia et al., 2024)	Best DNN/RNN-LSTM
Overall Accuracy	98.4%	97.0%	97.0% (Binary)
Attack Recall	100% (Injection & Flooding)	99.6%	93%
Macro F1-Score	0.984	0.52–0.93	0.92
Feature Count	13	13 / 76	13 / 76
Inference Speed	Microseconds	Milliseconds	Milliseconds
Interpretability	High	Low	Low

Table 2: Comparison with State-of-the-Art Deep Learning Models

The XGBoost + SMOTE model outperforms all deep learning baselines in overall accuracy, attack recall, and Macro F1-Score, while requiring significantly fewer computational resources and offering superior interpretability — critical qualities for real-world deployment in resource-constrained WSN environments.

7. CONCLUSION

This work successfully demonstrates the design and implementation of a high-performance, machine learning-based Intrusion Detection System (IDS) for Wireless Sensor Networks. By integrating aggressive feature reduction (154 → 13 behavioral features), SMOTE-based class balancing, and XGBoost classification, the system achieves state-of-the-art performance in real-time multiclass intrusion detection.

The SMOTE preprocessing strategy effectively resolves the severe 91% class imbalance inherent in the AWID dataset, enabling the model to detect all attack vectors — including rare injection and impersonation attacks — with near-perfect recall. The XGBoost model, configured with 150 trees and maximum depth of 10, delivers 98.4% overall accuracy and a Macro F1-Score of 0.984, outperforming CNN, DNN, and RNN-LSTM baselines from recent literature.

The feature correlation analysis mathematically validates the 13-feature selection, confirming that the chosen indicators are non-redundant and information-rich. This reduction not only improves computational efficiency for real-time edge inference but also enhances model interpretability — a crucial requirement for practically deployable security systems.

Overall, the proposed framework offers a scalable, reliable, and practically deployable solution for securing wireless sensor networks, with direct applicability to smart city infrastructure, healthcare IoT, industrial monitoring, and precision agriculture deployments.

8. REFERENCES

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